Trail Balance

Planning a lighter hiking backpack

Linear Programming · First progress presentation

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The packing problem

What should we pack for this hike?

Essential equipment and supplies
Food and water for the stated trip
A manageable backpack load

First example: fixed essentials and water, with two food quantities to choose

Project goal and first-milestone scope. Essential-supply categories: [R3, R4].

Our first hiking scenario

10 km
500 m ascent
4 hours assumed

One assumed 70 kg adult
Mild conditions, marked route
No overnight stay
No planned refill

The conditions define what the packing plan must cover.

Route, duration, conditions and body mass: classroom assumptions.

Basic supplies that stay in the pack

Essential group	Examples
Navigation and light	Phone, map, compass, headlamp, backup power
Weather protection	Warm layer, rain shell and sun protection
Emergency supplies	First aid, whistle and emergency shelter
Practical supplies	Containers, repair tools and waste bag
Food reserve	Separate emergency food supply

Fixed kit: 2.3 kg, including the backpack and packaging. Masses are assumed.

Essential categories: [R3] National Park Service and [R4] American Hiking Society. All masses: assumptions.

Water and carrying allowance

Water

2.5 L ≈ 2.5 kg

0.5 L/hour × 4 hours = 2 L
Add an assumed 0.5 L reserve

Pack allowance

7 kg

10% of the assumed 70 kg body mass
A general day-hiking guideline

Individual carrying limits need separate assessment.

Water guideline: [R4]. General pack-weight and fit guidance: [R5, R6]. Personal-capacity limitations: [R9, R10].

Food choices we can vary

Per serving	Tortilla	Tuna
Edible mass	50 g	74 g
Food energy	140 kcal	70 kcal
Protein	4 g	17 g
Available servings	6	2

The labels provide the coefficients. The available quantities are assumed.

Nutrition-label data: [R7] Mission Restaurant Style tortillas and [R8] StarKist 74 g tuna pouch.

What the packing plan must meet

Check	Requirement in this example
Ration energy	At least 630 kcal
Ration protein	At least 33 g
Available stock	At most 6 tortilla and 2 tuna servings
Whole backpack	At most 7 kg, including fixed supplies

The food targets define one classroom ration. They are chosen assumptions.

Ration targets, stock and kit masses: assumptions. Planning allowance uses [R5].

How linear programming fits

Decision: servings of each food

x_1 = tortilla servings x_2 = tuna servings

Minimize food mass: $50x_1 + 74x_2$ grams

Meet the food targets and respect available stock.
Check the total backpack allowance.

The professor’s simplex method finds the best feasible mix.

LP framework: [R1] course simplex handout and [R2] Matoušek & Gärtner, Understanding and Using Linear Programming.

Solving the example with simplex

Packing step	Ration mass
All available food	448 g
After the first pivot	339.2 g
After the second pivot	274 g

Each step keeps the ration feasible. The appendix shows the two pivot calculations.

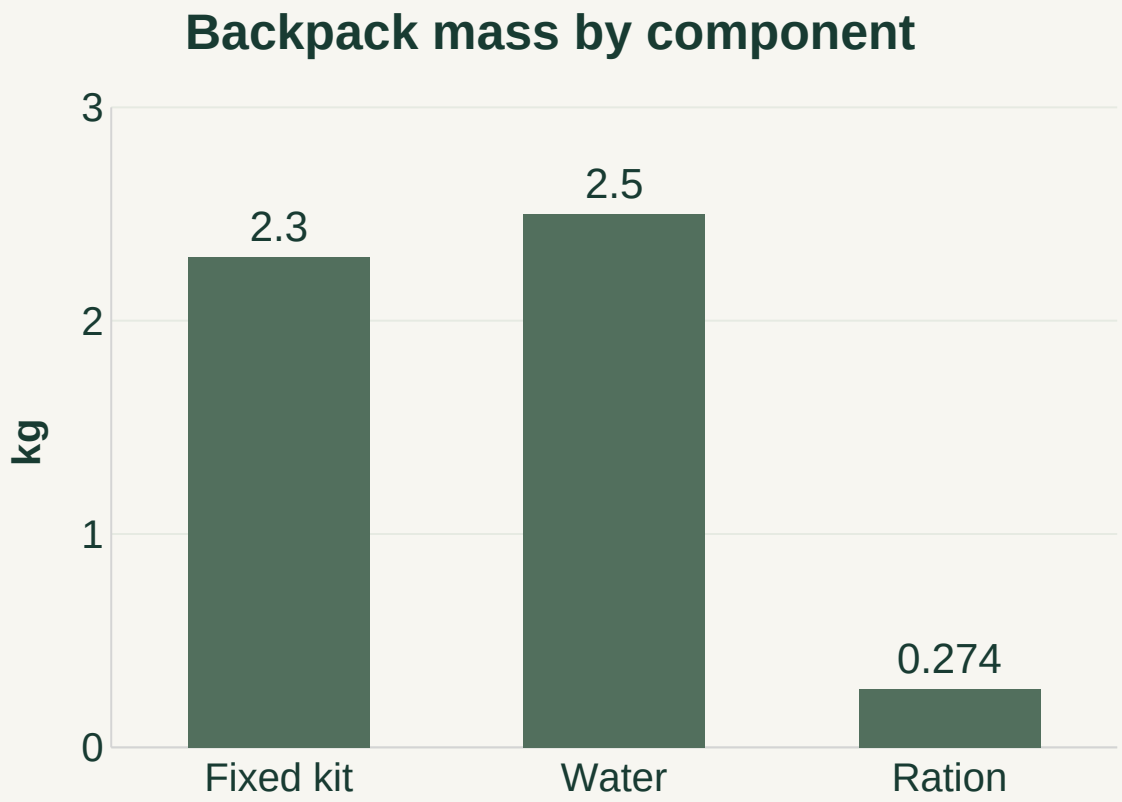
Exact project solution using the primal simplex method in [R1]. Intermediate mass rounded to 0.1 g.

The proposed packing list

4 tortilla servings
1 tuna serving

274 g of ration

630 kcal and 33 g protein
Whole backpack: 5.074 kg
Planning margin: 1.926 kg



Classroom model result. Food-label coefficients: [R7, R8]. Kit and water quantities are scenario assumptions.

What this first example establishes

A packing decision can be solved and checked against stated needs.

The food choice saves 174 g compared with packing all the stock.

This inventory fits comfortably within the chosen pack allowance.

Next data: measured kit masses, more foods and a tighter capacity scenario

Current kit masses and food targets are classroom assumptions.

Interpretation of the stated classroom model. Measurements and richer scenarios remain future work.

Future work and website

Expanding the packing problem

Measured masses and packed volume
More food and equipment choices
Whole items and shared supplies
Fair allocation across a group

Website direction

Trip details and available inventory
An understandable packing list
Total load and remaining capacity
Clear explanations when needs conflict

Each progress presentation adds one manageable extension.

Planned extensions. Whole-equipment selection, group allocation and website implementation remain future work.

Appendix: LP formulation

x_1 = tortilla servings x_2 = tuna servings

Minimize $M = 50x_1 + 74x_2$

$140x_1 + 70x_2 \geq 630$ kcal

$4x_1 + 17x_2 \geq 33$ g protein

$0 \leq x_1 \leq 6, \quad 0 \leq x_2 \leq 2$

$4800 + M \leq 7000$ g

Project formulation using [R1, R2]. Targets and equipment masses are assumptions.

Appendix: simplex reformulation

$y_1 = 6 - x_1 \qquad y_2 = 2 - x_2$

Maximize $\zeta = 50y_1 + 74y_2$

$140y_1 + 70y_2 \le 350$

$4y_1 + 17y_2 \le 25$

$0 \le y_1 \le 6, \qquad 0 \le y_2 \le 2$

Zero removal keeps the full inventory and gives a feasible slack basis.

Equivalent maximum-removal LP. Feasible slack start follows [R1], physical pp. 3–8.

Appendix: matrix and starting basis

$$x = (y_1, y_2, s_1, s_2, s_3, s_4)$$

A	y_1	y_2	s_1	s_2	s_3	s_4	b
Energy	140	70	1	0	0	0	350
Protein	4	17	0	1	0	0	25
Tortilla stock	1	0	0	0	1	0	6
Tuna stock	0	1	0	0	0	1	2

c = (50, 74, 0, 0, 0, 0). Initial basis matrix: I_4.

Matrix and slack-variable notation adapted from [R1].

Appendix: simplex pivot 1

Reduced costs (−50, −74). Enter y_2.

Basic	Value	Direction	Ratio
s_1	350	70	5
s_2	25	17	25/17 (minimum)
s_3	6	0	Excluded
s_4	2	1	2

Step = 25/17. Protein slack s_2 leaves the basis.

Exact arithmetic using [R1], primal simplex steps. Slacks $s_1 \ldots s_4$ follow the preceding matrix.

Appendix: simplex pivot 2

Improving reduced cost: −554/17. Enter y_1.

Basic	Value	Direction	Ratio
s_1	4200/17	2100/17	2 (minimum)
y_2	25/17	4/17	25/4
s_3	6	1	6
s_4	9/17	−4/17	Excluded

Step = 2. Energy slack s_1 leaves. Final removal: y = (2, 1).

Exact project arithmetic using [R1]. Final reduced costs confirm optimality.

Appendix: optimality certificate

$\lambda_1 = 277/1050$ $\lambda_2 = 49/15$ both ≥ 0

$140\lambda_1 + 4\lambda_2 = 50$

$70\lambda_1 + 17\lambda_2 = 74$

Every feasible ration has $M \geq 630\lambda_1 + 33\lambda_2 = 274$ g.

Our feasible ration attains 274 g, proving the minimum.

Project certificate. Duality framework: [R2], Chapter 6.

References: LP and hiking guidance

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[R2] Jiří Matoušek & Bernd Gärtner. Understanding and Using Linear Programming. Springer, 2007. doi:10.1007/978-3-540-30717-4

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[R4] American Hiking Society. The Ten Essentials of Hiking. americanhiking.org/10essentials/

[R5] Joe Pasteris, REI. How Much Should Your Pack Weigh? rei.com/learn/expert-advice/backpacking-weight.html

References: fit, foods and research

[R6] REI. How to Size and Fit a Backpack.
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[R7] Mission. Soft Taco Restaurant Style Flour Tortillas, label.
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[R8] StarKist. Light Tuna in Water (Pouch), 74 g label.
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[R9] Walsh, Low & Arkesteijn. Older-adult load-carriage study.
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[R10] Vickery-Howe et al. Male/female military load-carriage study.
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